Joint fixed sets and the remaining original prefix

Human Compression

Abstract

Two descriptions of the same joint fixed set can force every member of a string marking to preserve it. This gives an obstruction to generation that uses only the original permutation action. We prove the criterion and specify its application to the exact prefix of original job [68, 5]. The complete exploratory fixed sets agree and suggest a nine-point certificate. That exploration is retained as a diagnostic; a separately qualified producer and strict Saved check, explicit reuse of the target group's original generator lookup, and independent actual review remain required before job adoption.

1 The set criterion

Let an ambient group Γ act on a set Ω, and let $G \leq \Gamma$ be the target group, with right-action notation $x \mapsto x^g$. For $g \in \Gamma$, write $\mathrm{Fix}(g) = \{x \in \Omega : x^g = x\}$. A string marking of rank r is a sequence $(t_0, \ldots, t_{r-1})$ satisfying

$$[t_i, t_j] = 1 \quad \text{if } |i - j| \geq 2.$$

In the usual applications the marks are involutions. The following criterion does not require that additional assumption.

Lemma 1. *If g commutes with u and v, then g preserves $\mathrm{Fix}(u) \cap \mathrm{Fix}(v)$ setwise.*

Proof. For x fixed by u and v, commutation gives

$$(x^g)^u = (x^u)^g = x^g, \qquad (x^g)^v = (x^v)^g = x^g.$$

Thus the image of the joint fixed set under g is contained in that set. The same argument applies to g^{-1} and gives the reverse inclusion. This works for infinite as well as finite sets. □

Theorem 2. *Let the marks belong to Γ, and suppose $r \geq 5$. Put $a = t_0$, $b = t_1$ and $e = t_{r-1}$. If*

$$S = \mathrm{Fix}(a) \cap \mathrm{Fix}(e) = \mathrm{Fix}(a) \cap \mathrm{Fix}(b), \tag{1}$$

then every mark preserves S setwise. If there are an actual member $g \in G$ and a point $x \in S$ with $x^g \notin S$, then $H = \langle t_0, \ldots, t_{r-1} \rangle \neq G$. If the marks also belong to G, then $H < G$.

Proof. The two definitions of S show that a, b, e each fix it pointwise. For every index i with $2 \leq i \leq r - 3$, the string relations make t_i commute with both a and e. These marks preserve S by Lemma 1, using the first description in (1).

The only remaining index is $i = r - 2$. Since $r \geq 5$, its distance from indices 0 and 1 is at least two. The penultimate mark therefore commutes with a and b, so it preserves the second

description of S. This accounts for every mark, including the boundary case $r = 5$, where the middle range contains only $i = 2$.

The subgroup they generate lies in $\mathrm{Stab}_\Gamma(S)$. The displayed escape gives $g \notin H$, whereas $g \in G$. Consequently $H \neq G$, regardless of whether the marks belong to the target group. If their membership in G is separately known, then $H \leq G$ and hence $H < G$. □

We may take $\Gamma = \mathrm{Sym}(\Omega)$, so no original membership proof for the prescribed permutations is needed for the negative generation conclusion. The escape itself implies that S is nonempty and proper. Neither faithfulness nor transitivity of the action is used. No field, linear representation, characteristic-vector computation, order calculation or intersection property is needed. In particular, the theorem does not require a and b to commute.

The setwise-stabilizer escape principle is already used in the project's *Original invariant sets*, edition 2, `original_invariant_sets/v2/ARGUMENT.rst`. The new specialization here is the use of the two centralizer descriptions in (1) to cover every position of a string marking. Root proposed this equality criterion after the first branch 68 exploration. No claim of historical priority is made. The separate characteristic-vector and maximal-endpoint theorems remain useful other criteria; no old operational verdict is transferred to this new predicate.

2 The exact original prefix

The original job is $[68, 5]$. The exact whole pair file is `pair05.g`, 549385 bytes, with SHA256

`594ce69d6dd535c6178f87c9515805312fa6f0d69bb6a9bd89c48142643982ca`.

Its ordered entries are (a, e). The exact whole third-mark file is `branch0068.g`, 276591 bytes, with SHA256

`7aa368f3b42f704c1ea1e2047dc7844dd3215f78327584b547113ba994761723`.

Its value is b. The original end-to-end traversal stores the three values as (a, e, b) but places them at marking positions (t_0, t_{r-1}, t_1). Thus the rank-five prefix is

$$(a, b, _, _, e).$$

This order is essential: neither a source ordinal nor a third mark from another branch may replace this literal original b.

The target original group/model premise is the whole `q8_model.g`, 733634 bytes, SHA256

`5d08810d95f0c9454fbbd687133a23634e3a1c30c78b9c71cb4ea370cfa3cd0a`.

Its named-group and software meanings remain imported premises. The model defines the target variable G before the separate variable A, with model name `G2_8`. The target here is that original G, not A, the full automorphism group or its symplectic embedding. The accepted endpoint reversal replay `acd78878...0583c8d` supplies the exact original pair-membership limb for pair 05. The whole source, receipt, output and model joins of that acquired replay remain unchanged.

There is also an existing positive original-membership limb for this exact b. The native source `q8_high_adaptive2_branch0068.g`, 1987 bytes, SHA256

`57c508ee364c3c6761ca7ca92c756ee3b5a328dd5773279a3b03b18e0845b2c7`,

reads this whole pair and third mark. Before its later search, it explicitly rejects any of a, e, b outside the same original G or not of order two. Its receipt, 3274 bytes, `14022016...4047ac6`, records return code zero, the complete marker, empty standard error and unchanged inputs. Its complete output is 320 bytes, `91fa7c40...a546ce`.

This positive limb remains attributed historical context. The ambient form of Theorem 2 makes it unnecessary for the new negative argument. No new membership gate, native replay or pool import is required. A positive membership claim made for another purpose would require an explicit reuse acceptance retaining the whole subjects and software qualifications. The old empty leaf list and normalizer-pruning output are not used as a negative conclusion. The new scientific check must independently verify the whole permutation identities

$$a^2 = b^2 = e^2 = 1, \qquad [a, e] = [b, e] = 1.$$

It must *not* impose $[a, b] = 1$: the historical first bond has order four. These finite identities establish the prescribed involutory string prefix. They do not assert target-group membership, which is unnecessary for the exclusion.

3 The proposed finite certificate

The current exploration reports the complete sets

$$\mathrm{Fix}(a) \cap \mathrm{Fix}(e) = \mathrm{Fix}(a) \cap \mathrm{Fix}(b) = \{1, 10, 11, 12, 13, 14, 15, 16, 17\}. \tag{2}$$

It also reports the same triple intersection, with individual fixed counts $585, 73, 73$ for a, e, b. The first earlier scan established only the a, b set. The equality in (2) cannot be deduced from two equal cardinalities. It is the equality of the two complete sorted original point lists that is required.

The exploratory output has zero Source, proof and job credit. The qualified producer and strict Saved relation are to reconstruct both complete sets from the whole original inputs, rather than import (2) as an accepted finite result. Their certificate contains all three vector digests, all three fixed counts, the full AE, AB and triple lists, all required involution/commutation checks, and the supplied escape. A valid but inapplicable outcome must remain unpaid. Saved checks the supplied certificate directly; it does not call the producer or search for an escape.

An already acquired original ambient lookup is

$$10^{g_2} = 586, \tag{3}$$

where g_2 is original generator 2 of the target variable G in this same model. It is not a generator of A or an unspecified ambient automorphism. Both generator and point labels are *one-based*. This edge was actually checked by the preceding strict Saved profile computation, whose whole 4522-byte certificate has SHA256

`451c6dec5769465cfee483581f070e4a0084eb5ccb40dca32afe1b65f544768c`.

Its original generator meaning is bound to the accepted complete projective bridge, whose certificates have SHA256 `0601a736...184b4f8`, and to the whole bridge gate `4ecaf56b...3802648`. The current Root application acceptance `2ba1396a...e6c4aac3` binds that actual history and records the original one-based edge.

The bridge field `ordered_original_generator_images` retains the ordered original G generators. Its second entry is the target member used here. The lookup value (3) is independent of the endpoint pair: it is a value of that same original target permutation. Consequently it may be reused as an attributed lookup premise. The pair 08 fixed-set predicate from the preceding result is not reused for branch 68. The new finite check must establish that 10 belongs to the new common set and that 586 does not. A Root reuse gate must bind the exact original model, target G generator index, lookup, whole accepted bridge and whole producer/Saved history. No new matrix recovery or ambient-generator computation is needed.

Corollary 3. *Assume the exact original prefix-file/frontier ordering and target original G model/lookup premises above. If a qualified complete producer/Saved certificate establishes* (2)*, and the original lookup* (3) *is explicitly authenticated and reused, then no rank-five completion of original branch 68 generates G.*

Proof. The exact ordering of the original prefix makes a, b, e occupy the positions required by Theorem 2. Equality of the complete fixed sets gives its set S. The point 10 belongs to the displayed nine-set and 586 does not. The authenticated lookup therefore supplies an actual member $g_2 \in G$ moving a point of S outside S. The theorem proves the claim. □

This is a separate set argument for a fixed original prefix. It is not an application of the maximal-endpoint theorem to pair 05. That endpoint profile remains $(585, 73, 9)$ and outside the two rank-three-endpoint criterion. The present implication needs the actual original b, complete fixed-set equality and ambient escape.

4 Scope of any subsequent adoption

The current fixed required universe has 686 jobs. After the separately completed 76-job registration it has 685 closed jobs and the single pending job $[68, 5]$. The corollary would justify removing that one job only after the new Source, runtime, controls, complete producer/Saved, distinct actual review and typed Root application requirements have been met. Those are future operations; this document changes no state.

All old 685 proofs, all rank-four records, the original frontier and software qualifications, both protected progress files and unrelated state must remain intact. Closing the last job in this fixed universe would still not by itself establish global frontier exhaustion, the full automorphism/reversal census, a whole original-manuscript recovery, a conjecture result, numerical compression or Lean formalization. No such credit is claimed here.